

Computer Vision Practical Report:

HOMOGRAPHY MATRIX

1. AIM

To apply Homography transformation on an image using OpenCV to change its perspective view, display the original and transformed images, and save the result to disk.

2. PROCEDURE

- Load the image and select matching source and destination points.
- Calculate the 3×3 Homography matrix using `cv2.findHomography()`.
- Apply the transformation using `cv2.warpPerspective()`.
- Display the original and transformed images side-by-side using Matplotlib.
- Save the output image using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np
import os

# Read the input image
image_path = r"C:\Users\adminuser\Desktop\Python CV\E14-IMG.jpg"

image = cv2.imread(image_path)

# Check whether the image was loaded successfully
if image is None:
    print("ERROR: Image could not be loaded!")
    print("Check the file path:", image_path)
    exit()

print("Image loaded successfully.")
```

```
print("Image size:", image.shape)

# Four points from the original image
src_points = np.float32([
    [100, 100],
    [500, 100],
    [500, 400],
    [100, 400]
])

# Corresponding points in the transformed image
dst_points = np.float32([
    [50, 150],
    [550, 80],
    [500, 450],
    [100, 500]
])

# Calculate the Homography matrix
H, status = cv2.findHomography(src_points, dst_points)

print("\nHomography Matrix:")
print(H)

# Define output image size
width = 600
height = 600

# Apply Homography transformation
transformed_image = cv2.warpPerspective(
    image,
    H,
    (width, height)
)
```

```
# Display original image
cv2.imshow("Original Image", image)

# Display transformed image
cv2.imshow("Homography Transformed Image", transformed_image)

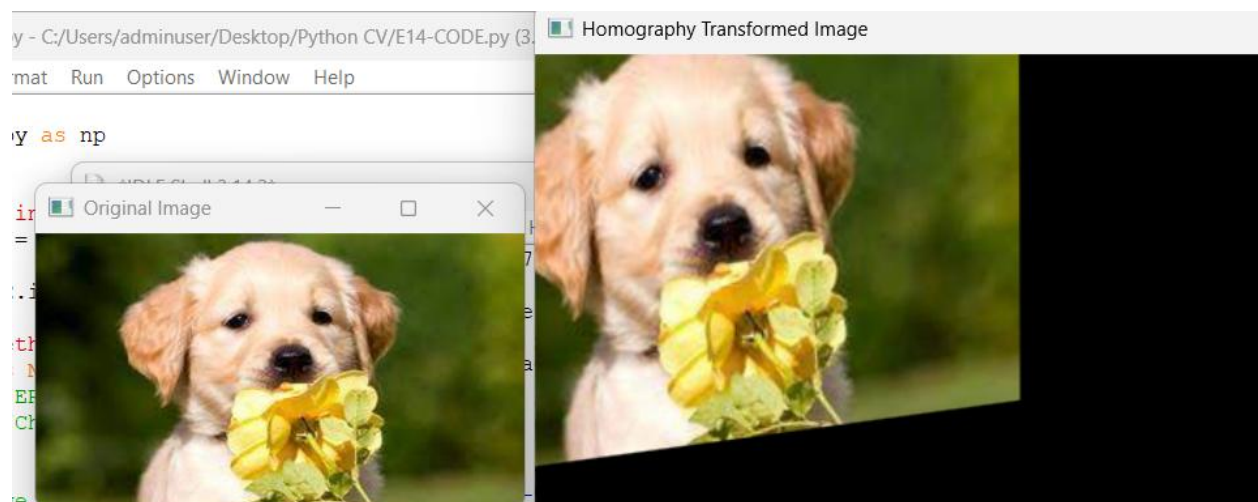
# Wait for a key press
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()

# Save transformed image
cv2.imwrite("transformed.jpg", transformed_image)

print("\nTransformed image saved as transformed.jpg")
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.